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MASTER THESIS

Addressing the Cold Start Problem in AI-Assisted Systematic Reviews

Start screening when previous research is scarce

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Abstract

Background

AI-assisted screening tools such as ASReview use active learning to prioritise relevant records, substantially reducing screening workload in systematic reviews. However, these tools require at least one known relevant record to initialise the model—a requirement that may not be met in time-sensitive settings such as crisis-driven rapid reviews. This "cold start" problem limits the applicability of AI-assisted screening when previous relevant literature is unavailable and only a limited number of records can be screened.

Objectives

To evaluate whether training the active learning model on a review's eligibility criteria, or on an LLM-generated title–abstract pair derived from those criteria, improves early screening performance relative to a cold start.

Methods

AI-assisted screening was simulated using ASReview across 23 labelled systematic review datasets from the SYNERGY repository (24,840 runs total). Four initialisation conditions were compared: cold start, eligibility criteria, LLM-generated abstract (GPT-4o-mini), and real relevant abstract. The outcome was the number of relevant records identified within the first 100 screened. An OLS regression with dataset fixed effects was fitted; confidence intervals and p-values were obtained via pairs bootstrap ($B = 10{,}000$).

Results

Training on eligibility criteria yielded significantly more relevant records than a cold start (mean difference = 11.07; $d = 0.54$; $p < .001$) and performed comparably to training on a real abstract. LLM-generated abstracts conferred no additional benefit over the criteria alone (mean difference = 0.17; $d = 0.01$; $p = .157$).

Conclusions

Eligibility criteria effectively resolve the cold start problem without requiring previous literature or LLM-based augmentation. As eligibility criteria are already produced in standard review workflows, this approach can be adopted immediately.

1 Introduction

1.1 AI-assisted systematic reviews

Systematic reviews are a cornerstone of evidence-based practice, providing comprehensive and unbiased syntheses of existing research on a given topic (Brignardello-Petersen, Santesso, and Guyatt 2025). However, systematically reviewing large bodies of literature can be time-consuming. In particular, screening titles and abstracts to identify relevant papers can be a major bottleneck limiting the practical applicability of systematic reviews (Bastian, Glasziou, and Chalmers 2010; Borah et al. 2017; Nussbaumer-Streit et al. 2021).

Fortunately, recent advances in machine learning have produced tools that assist researchers in screening papers by prioritizing the most relevant papers for screening first (Dos Reis et al. 2023; Ge et al. 2024; Ouzzani et al. 2016; Van De Schoot et al. 2021). One such tool is *ASReview*, an open-source software package for artificial intelligence (AI)-assisted systematic reviews (J. d. Bruin et al. 2025; Van De Schoot et al. 2021). ASReview uses active learning models to iteratively learn from the user's screening decisions and adaptively prioritise the most relevant papers for screening. Consequently, only a fraction of the papers returned by a systematic search need be screened to identify all the relevant ones (see Figure 1).

In short, AI-assisted screening is far more efficient than manual screening. Needless to say, there is always a risk that a paper that has not been screened would have been deemed relevant. However, this risk is offset by a decreased risk of relevant papers being missed due to human error caused by screening fatigue (Cai et al. 2025; Neeleman et al. 2026; Z. Wang et al. 2020). Moreover, the increased screening effiency allows for a larger search space. Taken together, AI-assisted screening increases the number of relevant papers that can be identified while reducing the time and effort required to conduct a review (Borah et al. 2017; Ferdinands et al. 2023; Nussbaumer-Streit et al. 2021; Schoot et al. 2025; Teijema et al. 2025).

1.2 The cold start problem

The active learning models used by ASReview adaptively prioritise the most relevant papers for screening based on the user's decisions. A training set of at least one relevant paper and one irrelevant paper is required by ASReview before the model can be trained. If no training set is provided, users will be presented with papers at random until at least one relevant and one irrelevant paper have been identified through manual screening (see Figure 1).

In practice, since only a fraction of the papers retrieved from a systematic search are relevant, the problem reduces to identifying the first *relevant* paper as quickly as possible. The most direct solution is to screen papers manually until a relevant one is found. However, if the prevalence of relevant papers is low, many papers may need to be screened before one is identified (see Table 1 for prevalence rates across the 23 systematic reviews used in this study).

Nonetheless, previous research has found that the number of papers that need to be screened to identify all relevant papers (or, in practice, until a stopping rule is triggered Boetje and Schoot 2024; Bron et al. 2025), is unaffected by what papers comprise the initial training set (Oude Wolcherink et al. 2023). This implies that screening at random until a relevant papers has been found is only a problem if a limited number of papers can be screened; for example, due to time pressure from supervisors or other stakeholders.

Since within a constrained screening window, every irrelevant paper screened without adaptive prioritisation represents a missed opportunity to identify additional relevant papers. We may thus define the scope of the problem more precisely:

> **Problem statement:** The cold start problem poses a practical barrier to AI-assisted screening when (1) no relevant papers are available before the start of screening, and (2) a limited number of papers can be screened.

Box 1. An applied example of the cold start problem.

During the corona crisis, timely decisions were crucial in mitigating the effects of the pandemic. Some rapid reviews reportedly took just 10 days, or even a few hours to complete (Tricco, Garritty, et al. 2020). Furthermore, what constituted as relevant continually shifted during the pandemic. At the start, for instance, experts had to rely on evidence from case studies and research into other pandemics and respiratory diseases (Brooks et al. 2020; Kurzer and Ornston 2023). Consequently, experts may have used outdated literature to train the AI model before screening, or spent valuable time consulting other experts to identify more recent literature. See subsection 5.1 for a full discussion of these issues.

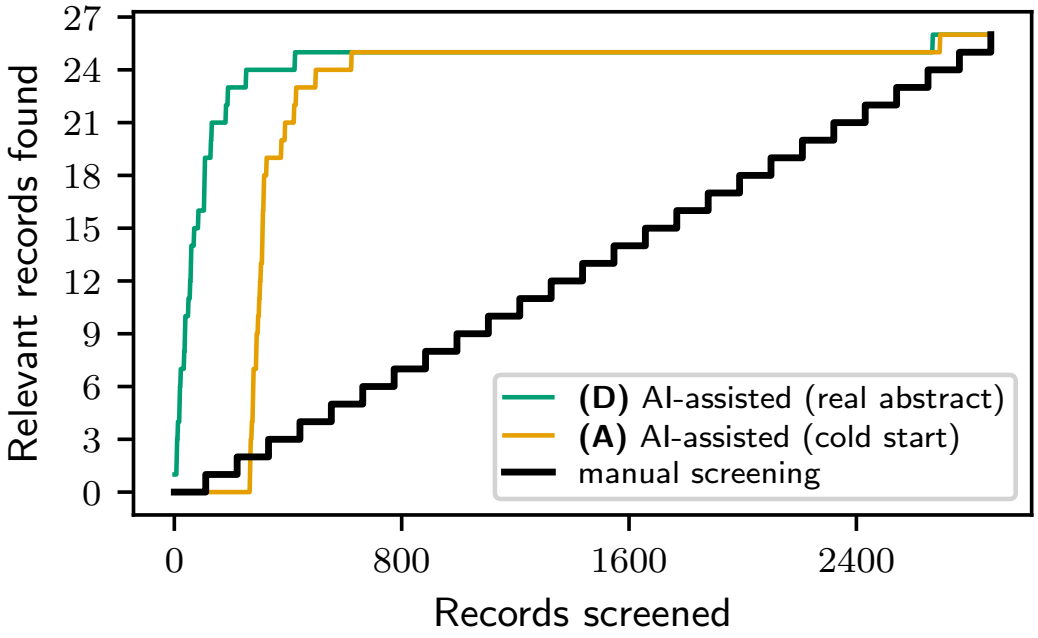


Figure 1: Recall curves for the simulated screening of *Appenzeller-Herzog_2019*, comparing AI-assisted screening initialised with a real abstract, AI-assisted screening with a cold start, and **manual screening**. The cold start curve begins to rise at the point on the x-axis where the first relevant paper is identified through manual screening. The recall curve for the real-abstract condition starts at $y = 1$ because the AI model is already trained on one paper before the first paper is screened. Note that the last relevant paper lacks an abstract and was therefore only retrieved after nearly all other papers had been screened.

1.3 Mitigating a cold start without relevant papers

If only a limited number of papers can be screened, manually screening until the first relevant paper is found is an insufficient solution to the cold start problem. A more effective approach may be to train the AI model on texts already available within the systematic review workflow. In particular, the **the eligibility criteria** defining the reasons for including or excluding a paper from a review may be used (Cochrane 2026).

Another possible solution to the cold start problem may be to use large language models (LLMs) to generate text to train the AI model on before screening. For instance, an LLM could be given the eligibility criteria and asked to produce a (possibly non-existent) title and abstract of a plausibly relevant paper. Training the AI model on this title-abstract pair before screening could result in a greater number of relevant papers being found within a limited number of papers screened (see Bachmann et al. 2025; W. Zhang et al. 2025 for examples of the use of LLMs to mitigate the cold start problem in other fields). However, LLMs are stochastic by nature, which makes it difficult to reproduce their results. Therefore, caution should be exercised when applying them to standard systematic review workflows (Gartlehner et al. 2025).

1.4 Objectives

This study aims to address the 'cold start' problem in AI-assisted systematic reviews. Specifically, we will investigate whether screening performance can be improved by training the AI model on a systematic review's eligibility criteria, compared to a cold start. Secondly, we will investigate whether screening performance can be further improved by training the AI model on a title and abstract generated by an LLM based on the eligibility criteria.

2 Methodology

To address the research question above, we simulated AI-assisted screening of published systematic reviews using ASReview. Specifically, we used 23 systematic reviews from the SYNERGY dataset (De Bruin et al. 2023). These datasets contain titles and abstracts of all papers screened by the original review authors, each labelled for relevance (0 or 1), enabling simulation of screening decisions with ASReview (see Byrne et al. 2024; Ferdinands et al. 2023; Teijema et al. 2025 for comparable simulation studies). Dataset metadata are summarised in Table 1. Across all simulations, we investigated the effect of four different starting conditions on the number of relevant papers found within the first 100 screened.

2.1 Simulation design

AI-assisted screening was simulated using ASReview's Python API (ASReviewLAB 2026). All code is available on GitHub (Van Ommeren 2025). Four ways of training the AI model before the start of screening were compared:

A. *Cold start* — no training beforehand; screening is manual until the first relevant paper is randomly found.

B. *Eligibility criteria* — ASReview is trained on the inclusion and exclusion criteria of the systematic review. (See Appendix B for the eligibility critera of Menon_2022.)

C. *LLM-generated* — ASReview is trained on a title and abstract of a plausibly relevant paper generated by an LLM based on the eligibility criteria (OpenAI's gpt-4o-mini model is used and is instructed via a DSPY module for dynamic prompt generation in each run (Khattab et al. 2023)). A new title-abstract pair is generated in each simulation run. (See Appendix B for an example title-abstract pair generated for Menon_2022.)

D. *Real abstract* — ASReview is trained on the abstract of a real paper from the set of all relevant papers (sampled with replacement between runs).

The text to train the AI model on in Condition B, C and D was provided as an example of a *relevant* abstract only. So, since ASReview requires at least one relevant abstract and one irrelevant abstract to prioritize papers by relevance, screening technically proceeded randomly until the first *irrelevant* paper was identified. In practice however the first paper screened is almost always irrelevant, so ASReview could be trained and adaptive prioritisation could effectively begin from the second paper onwards. The ELAS U4 configuration of ASReview was then used across conditions (see J. d. Bruin et al. 2025 for details).

The four conditions were compared on the number of relevant papers found within the first 100 papers screened. Screening was simulated 270 times per condition per systematic review (1,080 runs per dataset; 24,840 total).

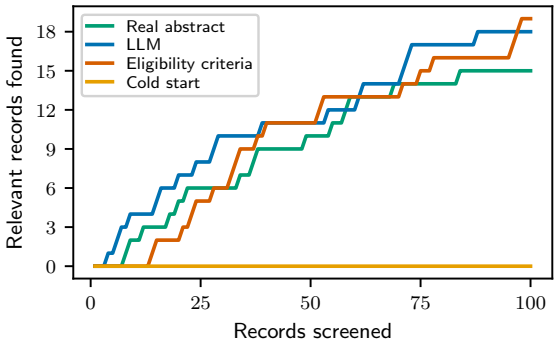
2.2 Analysis plan

For each systematic review dataset, recall plots were created to compare the four conditions, showing the number of relevant papers found by the number of papers screened. To visually interpret performance between runs as well as the average performance per condition; the recall curves of each individual simulation run are plotted at 7% opacity alongside the mean recall curve for each condition (bold dashed line). A spaghetti plot summarises the mean number of relevant papers found per condition across all 23 datasets.

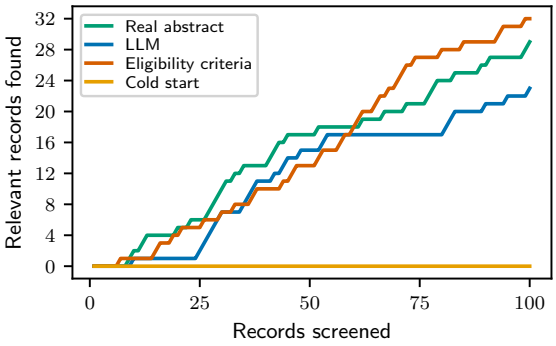
The outcome of each simulation run is the number of relevant papers found within the first 100 papers screened. An OLS regression was fitted with two predictors: starting condition (no-intercept coding, so each coefficient directly estimates the covariate-adjusted group mean), and a fixed effect control for baseline differences between systematic review datasets. OLS was chosen for interpretability. Confidence intervals and non-parametric p-values were obtained via a pairs bootstrap ($B = 10{,}000$ resamples).

Model-estimated condition means are plotted with 95% bootstrapped confidence intervals. Two pre-specified contrasts were examined: (1) cold start versus eligibility criteria, and (2) eligibility criteria versus LLM-generated title-abstract pair. The analysis was conducted in R (version 4.4.2), with the exception of the recall plots, which were created using matplotlib in Python.

(A) Example individual simulation runs

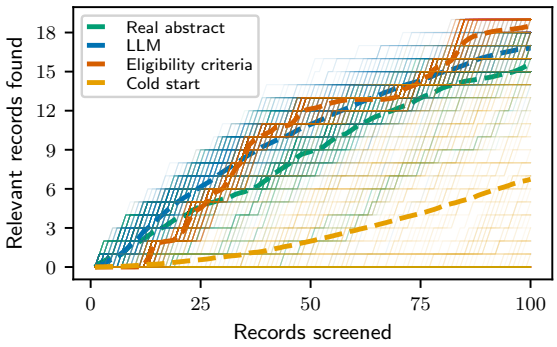


(i) Appenzeller-Herzog (2019)

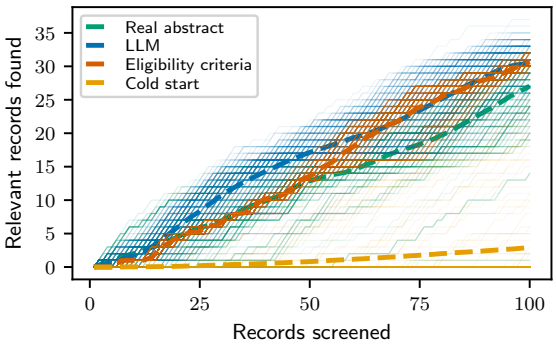


(ii) Brouwer (2019)

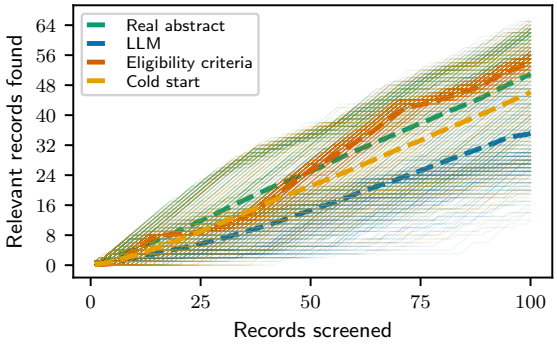
(B) Aggregate recall curves



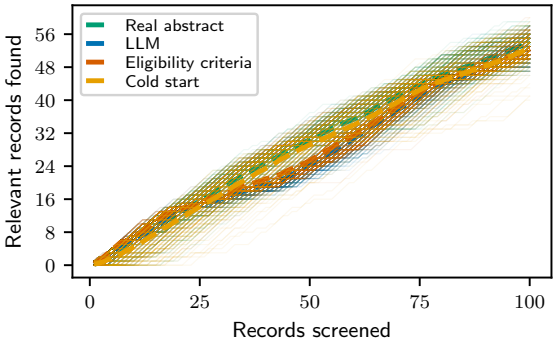
(iii) Appenzeller-Herzog (2019)



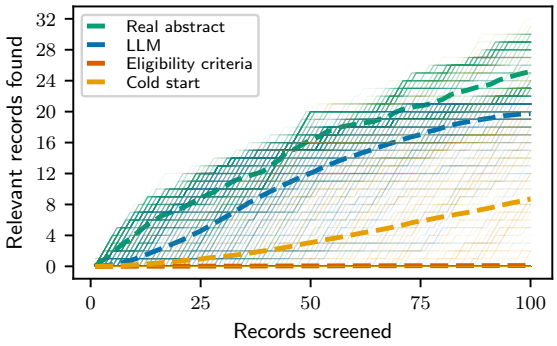
(iv) Brouwer (2019)



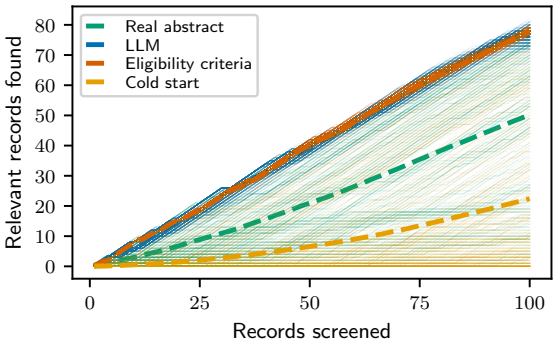
(v) Muthu (2021)



(vi) Nelson (2002)



(vii) Radjenovic (2013)



(viii) Walker (2018)

Figure 2: **(A)** Recall curves of exemplary individual simulation runs. **(B)** All simulations runs per conditions (thin lines), and average per condition (bold dashed lines). Colours indicate condition: cold start, eligibility criteria, LLM, and actual paper. Remaining datasets are shown in Appendix E.

3 Results

Simulation results are summarised below. More relevant papers were consistently found within the first 100 screened in the eligibility criteria condition, the LLM condition, and the real abstract condition than in the cold start condition (with a few exceptions, see Figure 3a). The difference between the cold start condition and the other three conditions widened as the percentage of relevant papers decreased: datasets with a high percentage of relevant papers showed little difference between conditions, while datasets with a low percentage showed the largest gain for the other three conditions over the cold start condition.

The cold start condition and the real abstract condition showed the highest between-run variability (see recall plots of all simulations runs per condition as well as their average in Figure 2 and Appendix E). Some runs in the real abstract condition led to markedly fewer papers found than the condition mean (e.g., 15 instead of 25 for Brouwer_2019).

Model-estimated means after controlling for dataset (95% bootstrapped CIs) were: cold start, $M = 20.45$ [20.09, 20.80]; eligibility criteria, $M = 31.51$ [31.35, 31.68]; LLM, $M = 31.68$ [31.51, 31.85]; real abstract, $M = 31.21$ [31.02, 31.41] (Figure 3b).

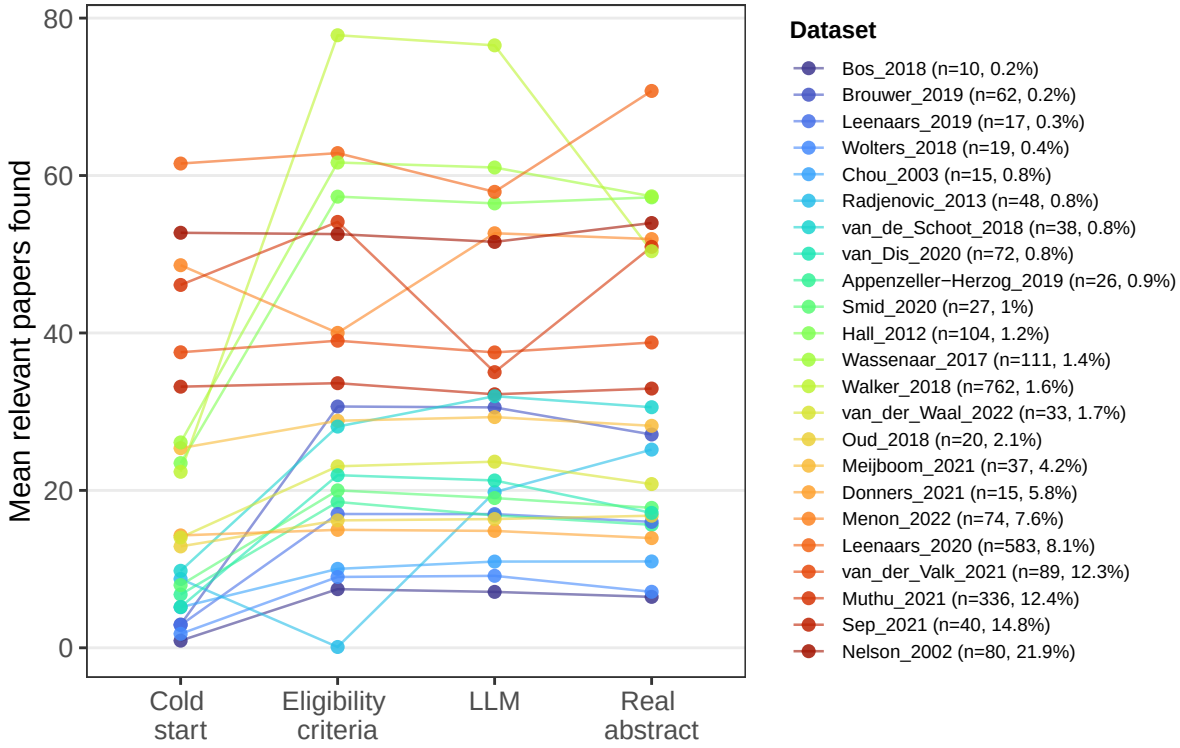
The eligibility criteria condition found significantly more relevant papers (11.07) on average than the cold start condition(95% CI [10.68, 11.45], $p < .001$; Cohen's $d = 0.54$). The LLM condition found 0.17 more papers than the eligibility criteria condition, a negligible and non-significant difference (95% CI [-0.07, 0.40], $p = 0.157$; $d = 0.01$).

4 Discussion

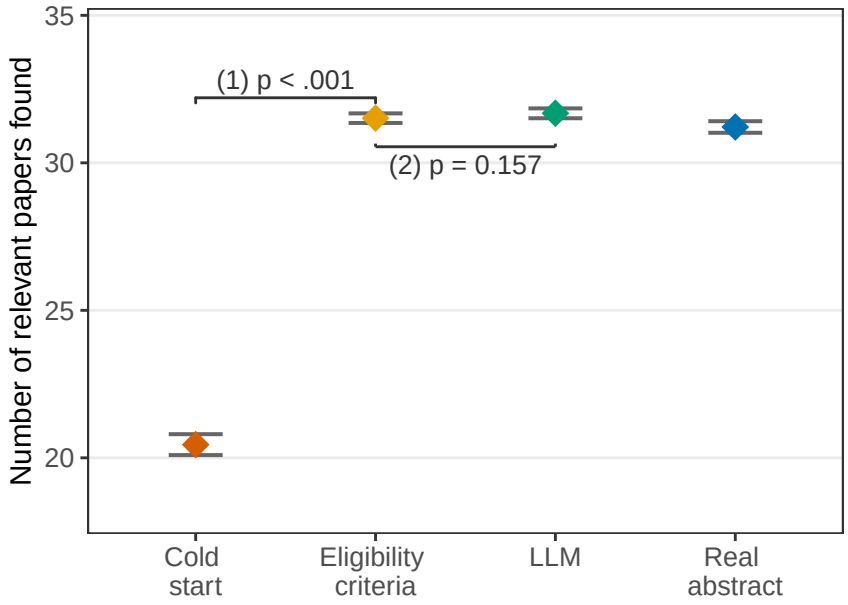
This study examined whether two strategies for training AI-assisted screening tools before the start of screening could outperform a cold start in identifying relevant papers within a limited number of papers screened. The two strategies were: training ASReview on a systematic review's eligibility criteria; and training ASReview on an LLM-generated title-abstract pair based on those criteria. As both strategies avoid the need for previous relevant literature, they are applicable in crisis settings such as the COVID-19 pandemic, where relevant examples may be unavailable or unrepresentative (see the practical implications below).

The first comparison of interest was found to be *significant*: training on the eligibility criteria yielded significantly more relevant papers than a cold start, with a mean difference of 11.07 papers per 100 screened ($d = 0.54$, $p < .001$). Notably, condition B on average performed comparably to training on a real abstract, though substantial between-dataset variability was observed. A supplementary analysis testing whether some eligibility criteria yielded more relevant papers than others due the specificity of the language used in those criteria (by expanding abbreviations or adding MeSH terms) found no consistent benefit of these alterations (Appendix C).

The first comparison of interest was found to be *insignificant*: LLM-generated title-abstract pairs provided no statistically significant benefit over the eligibility criteria alone (mean difference = 0.17 papers; $d = 0.01$, $p = .157$). This suggests that the LLM does not add information about relevant-paper characteristics beyond what is already encoded in the eligibility criteria, at least for the LLM and prompting strategy tested here.



(a) Mean number of relevant papers found within 100 papers screened per dataset across conditions. Each line represents one of the 23 systematic reviews. In the legend the total number and percentage of relevant papers in each dataset is shown. The datasets are ordered and coloured by percentage of relevant papers.



(b) Model-estimated condition means after controlling for dataset. Diamonds (♦) denote model-estimated means; error bars show 95% bootstrapped confidence intervals. Brackets indicate the two pairwise comparisons of primary interest.

Figure 3: The number of relevant papers found in the first 100 papers screened across the four conditions.

4.1 Limitations

Four limitations warrant acknowledgment. First, the simulations involved a potential data-leakage concern: because LLMs are trained on internet-scale text that may include the original SYNERGY papers, the LLM may have had implicit knowledge of the review content beyond what the eligibility criteria alone would convey. This could have artificially inflated the number of relevant papers found. Future work may consider using LLMs with known training cutoffs or apply the approach prospectively on new reviews.

Second, the LLM analysis was restricted to OpenAI'sgpt-4o-mini with minimal prompt engineering. This was due to early analyses finding no meaningful effect of prompt engineering (see Appendix D). Nonetheless, other LLMs or more elaborate prompting strategies may find different results.

Third, generalisability is constrained by the 23 SYNERGY datasets used. Future studies may replicate the results, for example, using the expected SYNERGY 2.0 dataset. Finally, the study was simulation-based. Empirical process analyses involving practitioners conducting reviews in information-scarce settings would strengthen ecological validity (Tricco, Zarin, et al. 2018).

5 Conclusion

This research shows that the cold start problem in AI-assisted systematic reviews can be effectively addressed by training the AI model on a review's eligibility criteria before screening begins. Training ASReview on an LLM-generated title-abstract pair derived from those criteria conferred no additional benefit.

The current approach could contribute to methodological developments in several ways. Firstly, the surprising similarity in effectiveness between the eligibility criteria and abstracts from real papers could inform the development of AI-assisted systematic reviews. Secondly, it could extend the applicability of AI-assisted screening tools such as ASReview to settings where information is scarce and time is limited. Ultimately, we hope that our research will inform evidence-based decision-making, particularly in crisis settings, thereby helping to make society more resilient as a whole.

5.1 Practical implications

Implication 1: Use eligibility criteria when previous relevant literature is unavailable. The eligibility criteria of a systematic review can serve as an effective substitute for previous relevant literature when initialising AI-assisted screening. This requires no additional effort beyond standard review workflows, in which eligibility criteria are already defined before screening begins (Cochrane 2026).

Implication 2: Avoid unsupervised LLM pipelines. The results caution against automated pipelines that apply LLMs without human oversight to generate training material for AI-assisted screening (Gartlehner et al. 2025). No benefit of LLM-generated abstracts was found. This is consistent with research finding that fully automated LLM-based search systems lack reproducibility and interpretability, while a human-in-the-loop approach produces more precise, contextually relevant queries (Vera et al. 2025).

Implication 3: Prefer eligibility criteria over a single known paper, even when one is available. Because training the AI model on the eligibility criteria (condition B) performed comparably to training on the abstract of a real paper (condition D),

we recommend using the eligibility criteria to initialise the AI model even when a single relevant paper is available. Doing so offers two advantages. First, the known paper can be held out as a verification check: if AI-assisted screening fails to retrieve it, this signals that screening may be incomplete, providing a principled stopping-rule criterion (Boetje and Schoot 2024).

Second, the available paper may not represent the most applicable literature. During the COVID-19 pandemic, for example, researchers relying on literature on previous pandemics and respiratory diseases to initialise the model may have identified an unrepresentative subset of relevant papers — a form of anchoring observed in other domains aswell (Rosbach et al. 2026). Although consulting domain experts remains the gold standard for identifying representative starting papers, this is not always feasible due to time, linguistic, or cultural constraints (Mikkola et al. 2024).

Acknowledgements

Thanks to all the people part of the AI-aided Knowledge Discovery Lab of Utrecht University and the employees of ForwardFaster for their feedback and contributions. A special thanks to Jelle Teijema and Timo van der Kuil for helping set up the simulation study. Finally, please note that various AI tools, such as DeepL, Github Copilot, ChatGPT and Claude were used to write Python and R code, and to improve the brevity and clarity of the written text. All original text was written by the authors.

Appendix

A SYNERGY metadata

Table 1: SYNERGY datasets overview

Nr	Dataset	Topic(s)	Records	Included	%
1	Appenzeller-Herzog_2019	Medicine	2873	26	0.9
2	Bos_2018	Medicine	4878	10	0.2
3	Brouwer_2019	Psychology, Medicine	38114	62	0.2
4	Chou_2003	Medicine	1908	15	0.8
5	Donners_2021	Medicine	258	15	5.8
6	Hall_2012	Computer science	8793	104	1.2
7	Leenaars_2019	Psychology, Chemistry, Medicine	5812	17	0.3
8	Leenaars_2020	Medicine	7216	583	8.1
9	Meijboom_2021	Medicine	882	37	4.2
10	Menon_2022	Medicine	975	74	7.6
11	Muthu_2021	Medicine	2719	336	12.4
12	Nelson_2002	Medicine	366	80	21.9
13	Oud_2018	Psychology, Medicine	952	20	2.1
14	Radjenovic_2013	Computer science	5935	48	0.8
15	Sep_2021	Psychology	271	40	14.8
16	Smid_2020	Computer science, Mathematics	2627	27	1.0
17	van_de_Schoot_2018	Psychology, Medicine	4544	38	0.8
18	van_der_Valk_2021	Medicine, Psychology	725	89	12.3
19	van_der_Waal_2022	Medicine	1970	33	1.7
20	van_Dis_2020	Psychology, Medicine	9128	72	0.8
21	Walker_2018	Biology, Medicine	48375	762	1.6
22	Wassenaar_2017	Medicine, Biology, Chemistry	7668	111	1.4
23	Wolters_2018	Medicine	4280	19	0.4

Note: Three of the datasets included in the original SYNERGY dataset were excluded entirely due to data quality issues: Chou_2004, Moran_2021 and Jeyaraman_2020.

B LLM generated abstracts

Eligibiltiy Criteria

To be eligible for inclusion in the SR sample, documents had to fulfil the following criteria:
1. identify explicitly as a "systematic review" in their title;
2. assess the effect of a non-acute, non-communicable, environmental exposure on a health outcome;
3. include studies in people or mammalian models;
4. be available in HTML format;
5. be published between 1 January 2019 and 30 June 2020.

LLM-generated Abstract

title: The Impact of Urban Air Pollution on Respiratory Health: A Systematic Review: **abstract:** This systematic review evaluates the effects of urban air pollution, a significant non-acute, non-communicable environmental exposure, on respiratory health outcomes in human populations. Spanning studies published between January 2019 and June 2020, this review incorporates findings from various research articles available in HTML format, ensuring compatibility with blinding strategies. The analysis details how exposure to pollutants, including particulate matter and nitrogen dioxide, correlates with increased incidences of asthma, chronic obstructive pulmonary disease (COPD), and other respiratory ailments. The review synthesizes data from multiple cohorts, emphasizing the need for public health interventions to mitigate pollution's impact on vulnerable populations. We discuss the implications of these findings in the context of urban planning and policy-making, aiming to foster environments that protect respiratory health.

C Supplementary analysis: effect of enriching the eligibility criteria

One possible explanation for why some eligibility criteria yielded more relevant papers than others is the specificity of the language used. As illustrated in Appendix B, some eligibility criteria contain abbreviations or generic terms that may not appear in the titles and abstracts of relevant papers. For instance, the eligibility criteria of Menon_2022 described the exposure of interest as "non-acute, non-communicable, environmental exposure" whereas the LLM substituted this with the concrete example "air pollution." This may explain why for some systematic review datasets, the use of their eligibility criteria led to more relevant papers found than others.

To test this directly, a supplementary analysis 'enriched' the eligibility criteria of 12 systematic review datasets (the top 12 in Table 1) in two ways: (1) writing out abbreviations (*Abbreviations*) and (2) expanding generic terms using MeSH vocabulary (*MeSH terms*). Each enriched version was compared to the original eligibility criteria using an OLS model with dataset fixed effects as covariate (4 replications per systematic review dataset for a total of 48 per condition and a grand total of 144 simulation runs).

Results are shown in Table 2. Neither enrichment led to a significant nor meaningful increase in the number of relevant papers found. The point estimates were small and in opposite directions (abbreviations written out: $\beta = -0.51$; MeSH terms added: $\beta = 0.56$), indicating no consistent effect of enriching the eligibility criteria. These results

suggest that the between-dataset variability observed in the main analysis is unlikely to be explained by the specificity of the vocabulary used in the eligibility criteria alone.

Table 2: Effect of enriching the eligibility criteria on the number of relevant papers found. Contrasts compare each enriched version to the original eligibility criteria.

Contrast (vs. original)	B	$SE^{\dagger}$	$p^{\dagger}$	95% CI†
Abbreviations written out	-0.50	0.31	0.198	[-1.20, 0.20]
MeSH terms added	0.56	0.31	0.134	[-0.14, 1.26]

† Standard errors, confidence intervals, and p-values should be interpreted with caution, since these assume homoscedastic residuals which might not hold.

D Supplementary analysis: effect of LLM parameter settings

The LLM condition involved three configurable parameters: the number of abstracts generated per prompt (1, 4, or 7), the instructed length of each abstract (100, 500, or 900 words), and the temperature setting of the LLM (0.0, 0.4, or 0.8). To test whether these settings affected screening performance, a supplementary OLS regression was fit on the LLM condition only (n = 6,210 runs), with dataset as fixed effect control as the main analysis. All three LLM parameters were mean-centred within the LLM condition before the analysis.

Results are shown in Table 3. Although all three predictors reached statistical significance, the effects are negligibly small in substantive terms. The direction of the effects is also difficult to interpret consistently. Most notably, a higher number of generated abstracts was associated with *fewer* relevant papers found ($\beta = -0.22$). The effect of temperature was positive ($\beta = 0.42$) but similarly trivial in magnitude. Taken together, these results suggest that the specific LLM parameter settings tested here had no practically meaningful impact on screening performance.

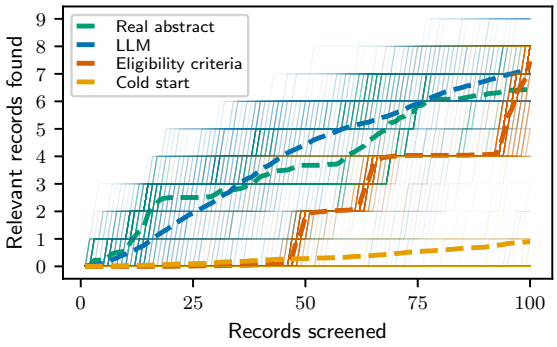
Table 3: Effects of LLM parameter settings on the number of relevant papers found (supplementary analysis, n = 6,210 runs). The model was fit on the LLM condition only, controlling for dataset and number of trials (OLS, Type II SS).

Parameter	B	$SE^{\dagger}$	$p^{\dagger}$	95% CI†
Number of abstracts	-0.22	0.02	$<.001$	[-0.25, -0.18]
Abstract length	0.03	0.01	0.041	[0.00, 0.05]
Temperature	0.42	0.14	0.002	[0.15, 0.69]

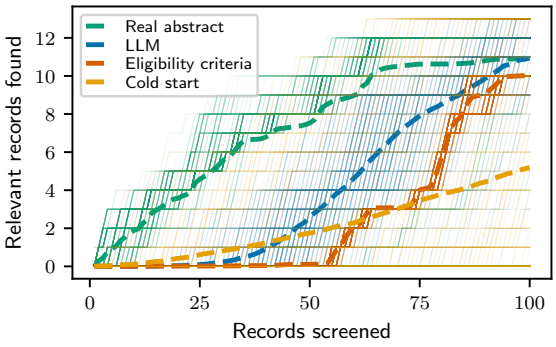
† Standard errors, confidence intervals, and p-values assume homoscedastic, independent residuals. Given the repeated-measures simulation structure and the heteroscedastic residuals identified in the main analysis, these should be interpreted with caution.

E Remaining aggregate recall plots

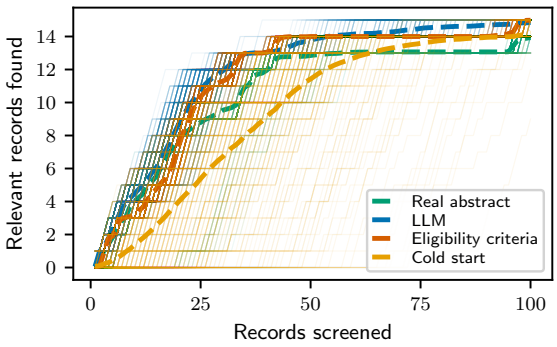
The following figures show the aggregate recall plots for the remaining 17 systematic reviews not shown in Figure 2. Line and colour conventions are identical to those in Figure 2.



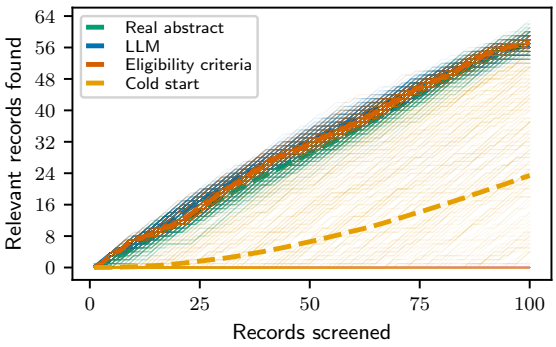
(a) Bos (2018)



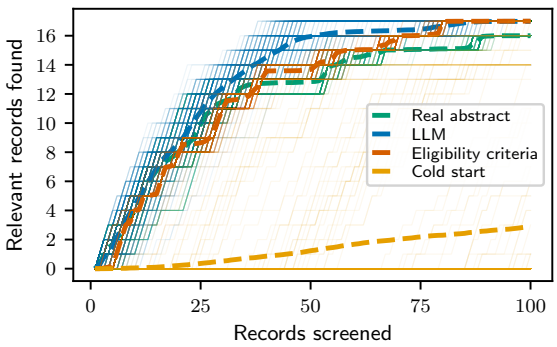
(b) Chou (2003)



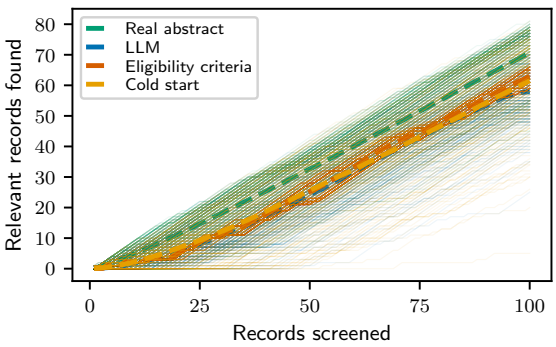
(c) Donners (2021)



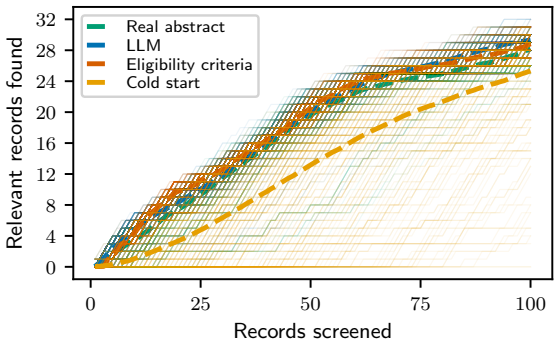
(d) Hall (2012)



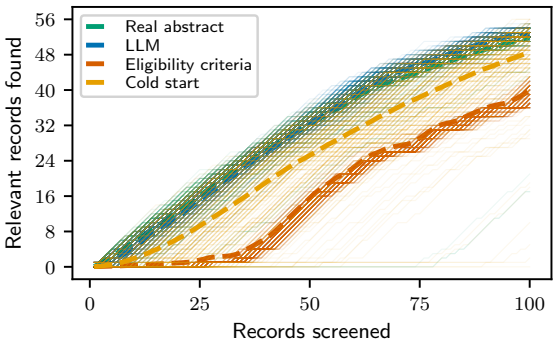
(e) Leenaars (2019)



(f) Leenaars (2020)

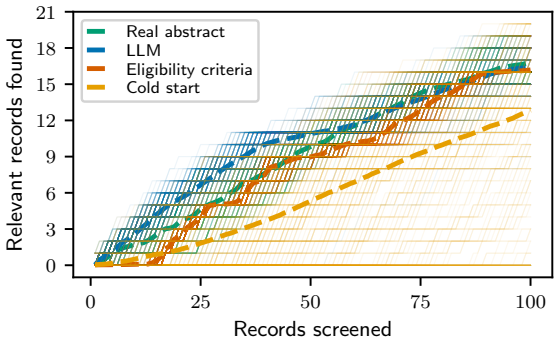


(g) Meijboom (2021)

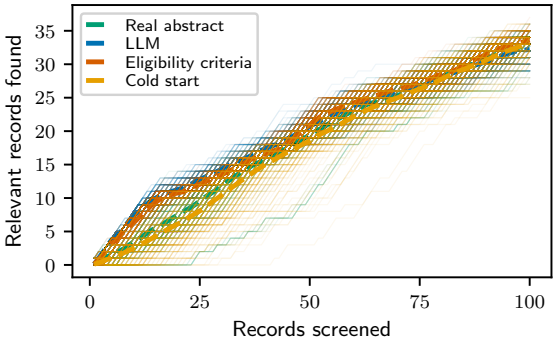


(h) Menon (2022)

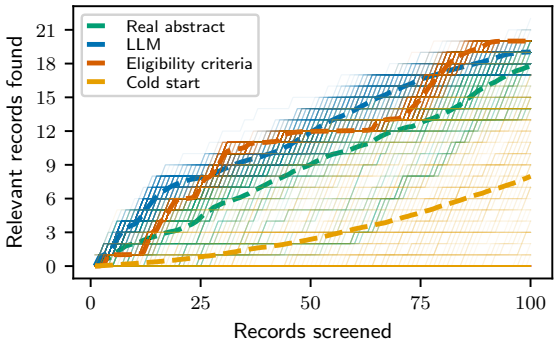
Figure 4: Aggregate recall plots for the remaining systematic reviews (1 of 3): Bos (2018) through Menon (2022). Thin lines show individual simulation runs; bold dashed lines show the condition mean. Colours indicate condition: cold start, eligibility criteria, LLM, and actual paper.



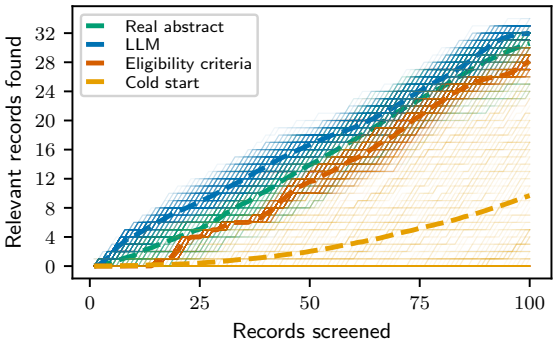
(a) Oud (2018)



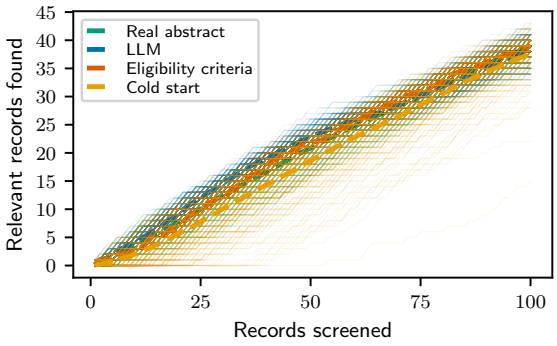
(b) Sep (2021)



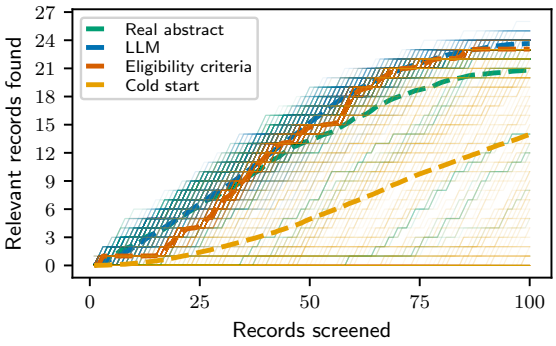
(c) Smid (2020)



(d) van de Schoot (2018)

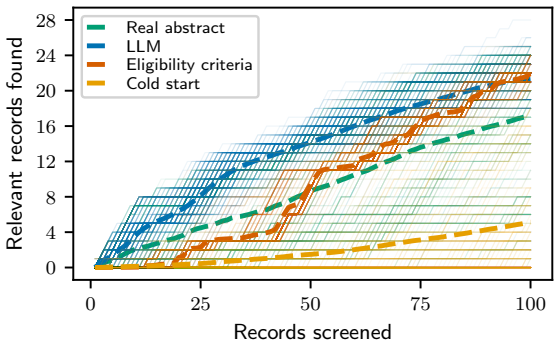


(e) van der Valk (2021)

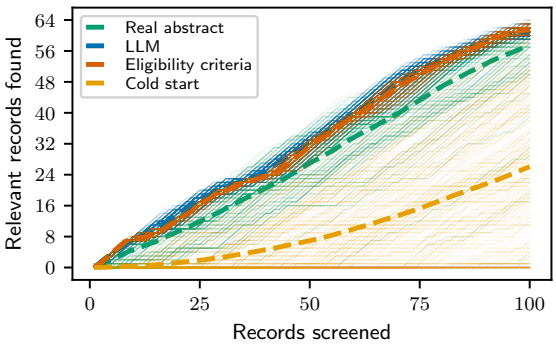


(f) van der Waal (2022)

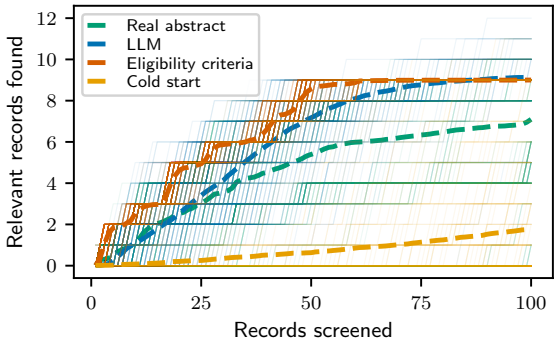
Figure 5: Aggregate recall plots for the remaining systematic reviews (2 of 3): Oud (2018) through van der Waal (2022). Thin lines show individual simulation runs; bold dashed lines show the condition mean. Colours indicate condition: cold start, eligibility criteria, LLM, and actual paper.



(a) van Dis (2020)



(b) Wassenaar (2017)



(c) Wolters (2018)

Figure 6: Aggregate recall plots for the remaining systematic reviews (3 of 3): van Dis (2020), Wassenaar (2017), and Wolters (2018). Thin lines show individual simulation runs; bold dashed lines show the condition mean. Colours indicate condition: cold start, eligibility criteria, LLM, and actual paper.

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